



MACHWORKS

VIRGINIA TECH



Preliminary Design Report

Camera Tracking

23 November, 2025

Chase B, Kent L, Nick M, Amogh S

[Insert Relevant Image (CAD/Simulation/Render)]



Authors

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Chase Bonfiglio	Project Manager	
Kent Lam	Operation Lead	
Nick Marrero	Technical Lead	
Amogh Saunshimath	Implementation Lead	



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Acronym List

MW – MachWorks

RF – Radio Frequency



Introduction

MachWork's Icarus mission is to develop and test a Mach 0.3+ high-performance aircraft using the Kentland Experimental Aerial Test Range. Clear footage enables post-flight analysis, provides real-time visual feedback during flight operations, and ensures that the status of deployment systems—such as the landing mechanism—can be reliably monitored. The Camera Tracking System is responsible for autonomously pointing a camera toward the aircraft throughout an entire mission. This requires precise motion control, sensors, and communication with both the primary ground station and an independent external onboard module.

To achieve these goals, the Camera Tracking System integrates a 3-axis gimbal, an XBee-based high-range telemetry link, a lightweight external tracking module, and a ROS2 software stack running on a Raspberry Pi 4B. These components create a closed-loop system capable of receiving position data, orienting automatically to maintain line-of-sight accuracy, and adjusting for processing delays using trajectory prediction. The final system must remain within strict mass, power, and budget limits while satisfying demanding accuracy and responsiveness requirements.

[Insert Figure: Conceptual diagram of full camera tracking setup]

[Insert Figure: Preliminary CAD sketch or block diagram]



Background

The Camera Tracking System gives freshmen an opportunity to go through an engineering project to better prepare them for integrating with one of MachWorks main teams. During a standard Icarus mission, operators rely heavily on continuous telemetry and visual feedback to ensure safe and informative flight tests.

The nominal operation of the Camera Tracking System is:

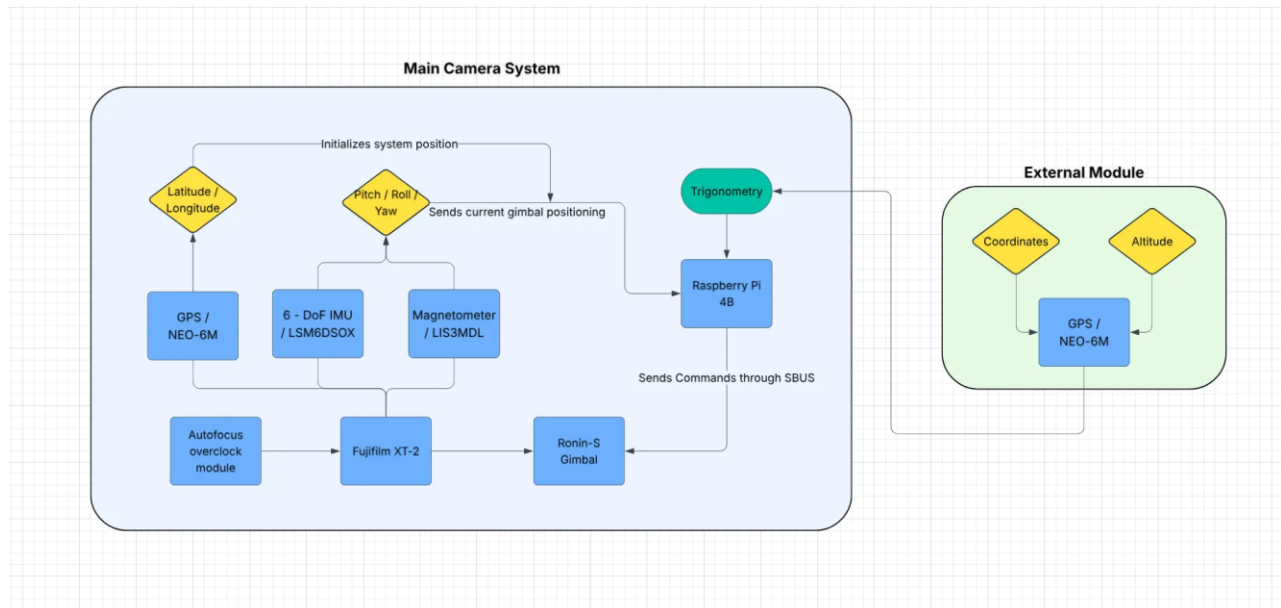
1. The ground station or external module transmits real-time GPS coordinates of the aircraft.
2. The ROS2 node on the Raspberry Pi 4B processes the incoming data.
3. Using orientation correction and trajectory prediction, the system computes the required azimuth and elevation angles.
4. The gimbal rotates to point the camera toward the expected location.

Because the system operates outdoors, physical constraints such as uneven terrain, wind, and high temperatures require the system to be capable of auto-correcting its orientation and maintaining stable tracking within $\pm 1^\circ$. The external module must remain small and lightweight to avoid disturbing the flight characteristics of any RC aircraft used for testing. All communication hardware must provide sufficient range, latency performance, and robustness in RF-dense environments.

Core equations that govern system behavior include spherical angle conversion from GPS coordinates, Kalman filtering to adjust for delay, and rigid-body transformations for gimbal orientation. Additional assumptions include typical GPS accuracy of ~ 2 m, magnetometer heading drift in the presence of iron structures, and the baseline performance characteristics of the DJI Ronin-S gimbal.



System Breakdown



The overall project is composed of two main subsystems. While the Main Camera System remains stationary near the user, the external module can be placed on the tracking subject as an alternative stream of telemetry data.

Main Camera Gimbal System	External Tracking Module
• DJI Ronin-S 3-axis gimbal • Fujifilm XT-2 • Raspberry Pi 4b • Hardware inverter Circuit • Digi XBee 3 PRO long-range transceiver • Triple-axis LIS3MDL magnetometer • Adafruit ISM33 • GY-NEO6MV2 • LiPo Battery • MAF94028: 4 Inch Omni Dipole Antenna	• ESP-WROOM-32 ESP32 ESP-32S Development Board • Digi XBee 3 PRO long-range transceiver • Triple-axis LIS3MDL magnetometer • LiPo battery • MAF94028: 4 Inch Omni Dipole Antenna • BMP390

Main Camera System



The Main Camera System serves as the primary sensing and pointing subsystem and remains co-located with the user during operation. It is built around a DJI Ronin-S 3-axis gimbal carrying a Fujifilm X-T2 camera, providing stabilized, high-quality visual tracking of the target. The Ronin-S offers high slew rates and a wide range of motion suitable for tracking fast-moving subjects, while the X-T2 delivers sufficient resolution and image quality for both qualitative assessment and potential post-processing or computer-vision tasks.

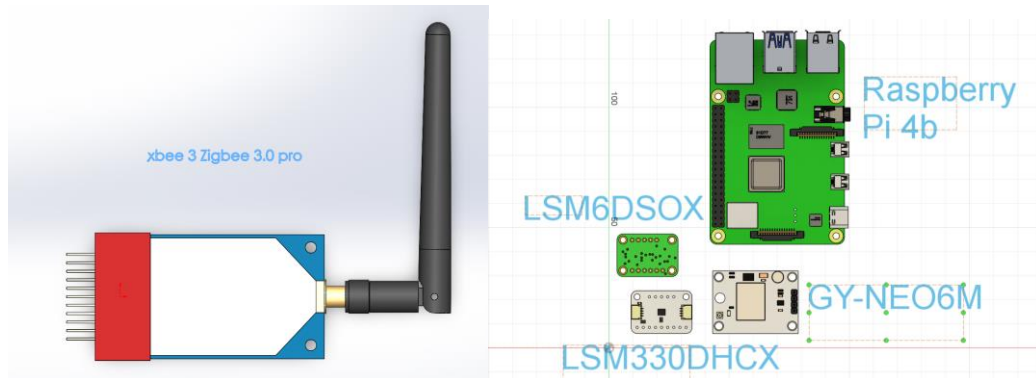
Control and communication functions are handled by an onboard Raspberry Pi 4B, which executes the tracking logic and interfaces with the gimbal. A hardware inverter circuit conditions signals as required by the gimbal control interface. Long-range, low-latency data exchange with the external tracking module and/or aircraft systems is provided by a Digi XBee 3 PRO transceiver, connected to a MAF94028 4-inch omni dipole antenna to achieve reliable coverage over the expected operating area.

To support attitude and position awareness on the main system, the payload integrates a triple-axis LIS3MDL magnetometer, an Adafruit ISM33 IMU, and a GY-NEO6MV2 GPS module. These sensors provide heading, orientation, and position data for the tracking algorithms and can be fused with external telemetry when available. All electronics are powered by a dedicated LiPo battery, sized to sustain the processing, communication, and sensing loads for the duration of a typical test, while keeping overall mass and form factor compatible with the gimbal and ground deployment constraints.

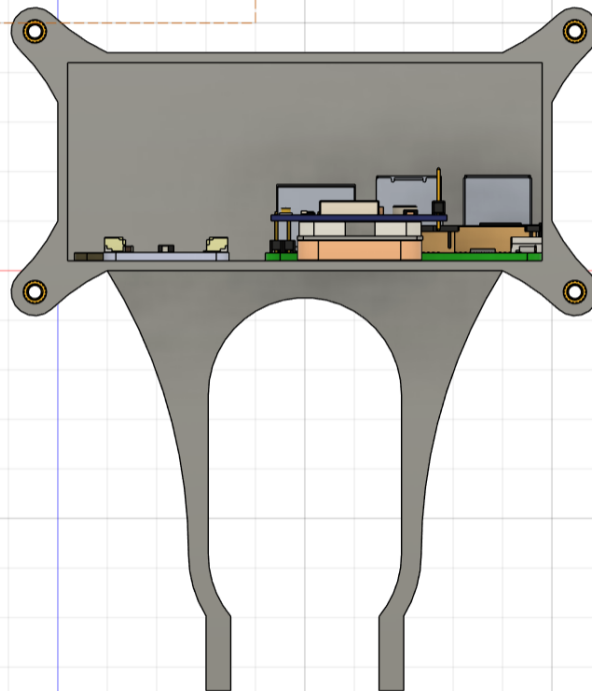
External Module

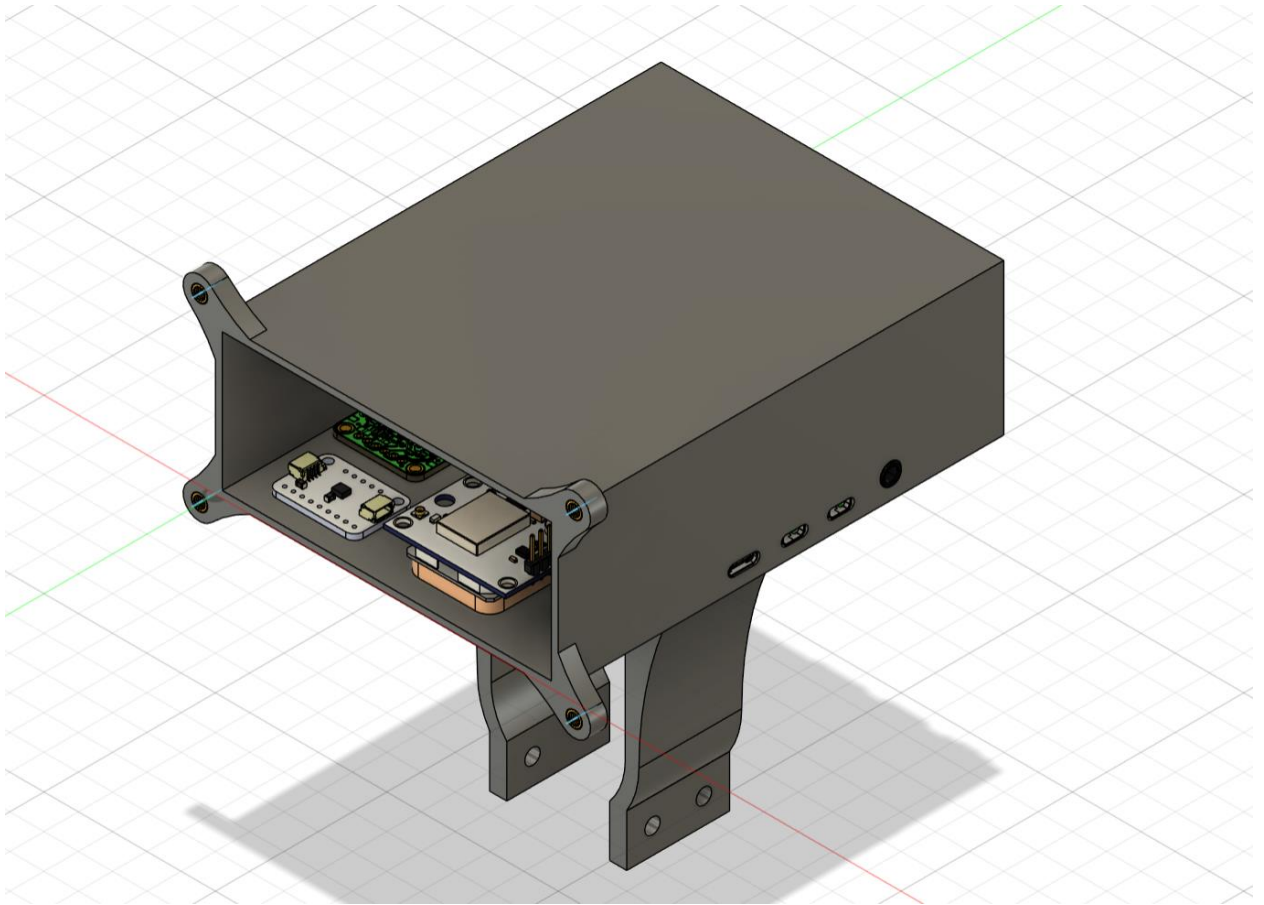
Kent add stuff here!

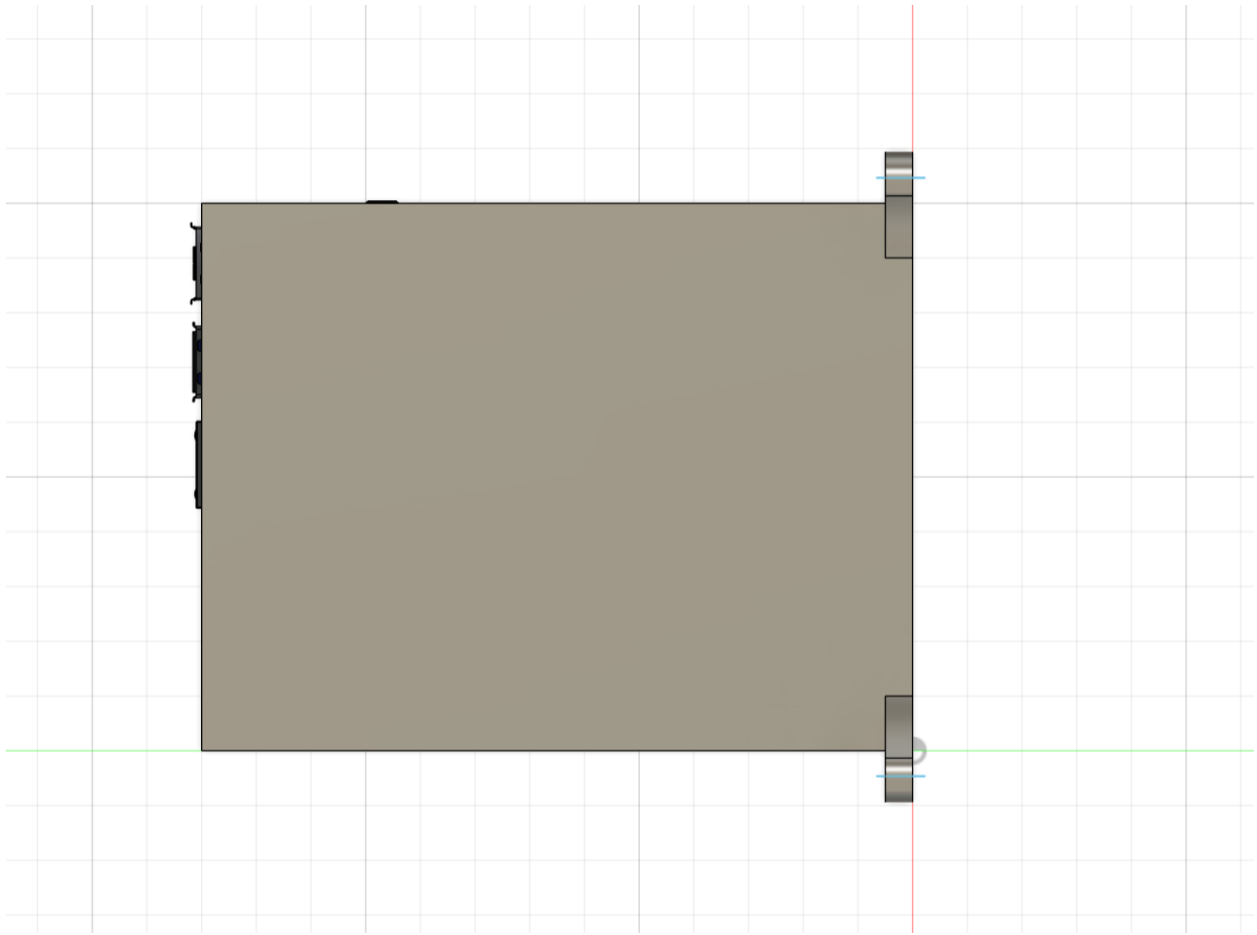
A 4-view CAD diagram will be added to depict placement of the gimbal, control electronics, and base structure. The full system weighs well under the required limits with the gimbal as the primary mass contributor.

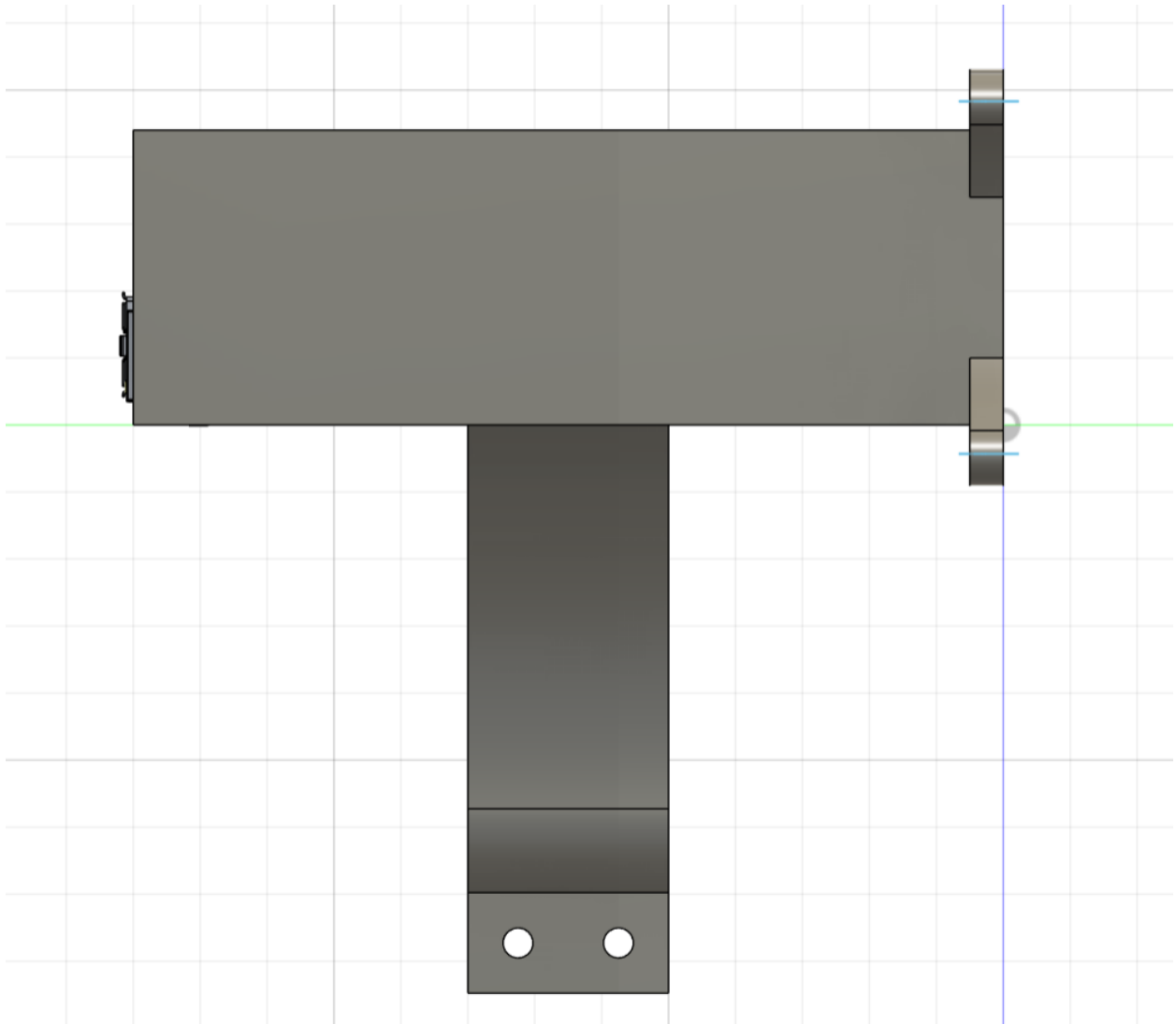


Camera Tracking Housing Module Assembly







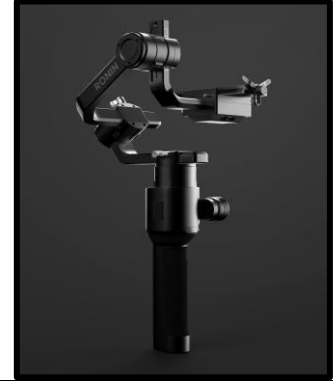




Overview of Components

DJI Ronin-S Gimbal

The Ronin-S is a 3-axis handheld gimbal capable of supporting payloads up to 2 lb—aligning directly with project requirements. It offers rapid slew rates suitable for tracking fast-moving aircraft and provides 360° continuous rotation capability when properly configured. The Ronin-S was selected over a star tracker due to its significantly higher rotational speed, full freedom of motion, and lower cost.



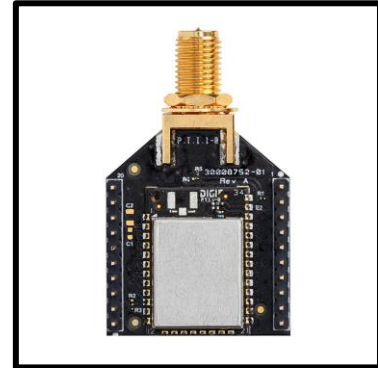
Criteria	Weight (%)	Gimbal	Grade	Star Tracker	Grade
Pointing Accuracy	15	High levels of control to capture footage of subject	4	Intended to orient it at high levels of camera zoom	5
Speed	30	High speed for following moving subjects	5	Much slower speed; designed to locate not track subject	2
Range of Motion	25	High ranges of motion	4	Medium range of motion; only designed to view sky	2
Weight Capacity	10	Works better with smaller rigs but designed to hold long range lens setups	4	Meant to work with long and large zoom lens setups	4
Cost	20	~\$200	4	~ \$300-\$600	1
Average Score			4.2		2.8



Weighted Score	4.2		2.45
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XBee 3 PRO Long-Range Transceivers

These transceivers provide up to 2 km of effective communication range with <10 ms latency and plug-and-play UART compatibility. The XBee was selected over LoRa due to lower latency (critical for real-time tracking, higher throughput, simpler integration and better mesh networking support.



Criteria	Weight (%)	Xbee 3 pro	Grade/10	LoRa	Grade
Range	25	2 km	8	.5km	5
Throughput & Latency	25	<10ms	7	200ms	3
Integration	20	Plug-and-play with UART	9	Manual register setup, driver code, and tuning	5
Power Consumption	5	45mA	5	120mA	3
Network	10	Full mesh	8	Point-to-point	5
Cost	15	~ 20	5	~ 20	5
Average Score			7		4.3
Weighted Score			7.35		4.4



ESP-WROOM-32

Serves as the microcontroller for the external module. It manages GPS acquisition, IMU orientation data and XBee communication.

LIS2MDL Magnetometer

A low-power 3-axis magnetometer used for heading correction on the external module. While sensitive to interference from ferromagnetic materials, careful placement and calibration mitigate these issues. It significantly improves heading estimation, especially when combined with GPS data.

External Tracking Module Enclosure

A lightweight 3D-printed enclosure will protect electronics while maintaining a total mass under 150 g. Non-PLA, heat-resistant materials will be used to ensure operation at 80°F.

MAF94028: 4 Inch Omni Dipole Antenna

A flexible antenna with wave coaxial dipole design for reduced ground dependence. 1.3dBi Gain @ 2.45 GHz



Software Overview

The flight-test payload requires a software stack capable of ingesting noisy telemetry, estimating the relative geometry between the aircraft and the ground station, and commanding the gimbal so that the onboard camera remains pointed at the desired target throughout high-speed maneuvering. The software is planned as a ROS 2–based system running on a single onboard computer (a Raspberry Pi or potential Nvidia Jetson), organized into several modular nodes rather than a single application.

At a high level, the architecture is divided into four main functional blocks:

Telemetry and Sensor Interface

A telemetry interface node will subscribe to aircraft state data (e.g., GPS position, altitude, heading, and potentially airspeed) supplied by the MachWorks avionics and/or ground system. In parallel, an on-gimbal sensor node will read local measurements, including GPS, IMU/compass data, and any gimbal feedback (encoders or internal status messages). Together, these data streams will be fused or transformed into a consistent reference frame to provide an estimate of both the aircraft and payload pose. This combined state estimate forms the basis for all downstream tracking and control logic.

Target Geometry and Control Logic

A central tracking node will compute the line-of-sight vector from the payload to the target and determine the required gimbal pan and tilt angles. This involves transforming the aircraft and payload positions and orientations into a common frame and solving the desired pointing direction. A control layer will then compare the desired angles to the current gimbal orientation and generate appropriate command signals. This controller will incorporate rate limits, mechanical constraints, and filtering to ensure smooth motion, avoid saturation, and reduce the impact of sensor noise and high-frequency disturbances.

Gimbal Command and Safety Layer

A dedicated gimbal interface node will manage all low-level communications with the gimbal via SBUS. This node will handle converting signals from the target geometry and control logic layer into commands appropriate for the camera gimbal.



Operator Dashboard and Data Logging

An operator-facing dashboard (e.g., a ROS 2–connected GUI) will provide real-time visualization of key system variables, including aircraft position, gimbal angles, tracking error, and system health/status flags. The interface will expose controls for switching modes (auto/manual), initiating safe shutdown, and adjusting a small set of operational parameters as needed during testing.

In parallel, a logging node will record telemetry, gimbal commands, and error/status messages to rosbags. These logs will support post-flight analysis, controller tuning, and debugging, as well as replay-based testing of the tracking algorithms.

Configuration and Development Approach

System parameters required for gimbal calibration, and telemetry update rates will be held in a configuration file. This approach allows rapid tuning and reconfiguration without code changes.

Telemetry inputs such as GPS and IMU data will be simulated in preliminary testing, in which the tracking and control nodes can operate on simulated telemetry instead of live data. This will enable iterative development and validation of the pointing and control logic in a lab environment prior to full hardware integration in the field.



Risk Analysis

The Camera Tracking System faces several operational and environmental risks. High-level risks include overheating, communication failure, and sensor drifting. These are mitigated through robust telemetry hardware, sensor fusion, proper thermal design, and careful calibration.

Risk Table

Risk	#	Severity	Likelihood	Notes
Overheating	1	Severe	Possible	Sun exposure or electrical components
Transmission loss	2	Severe	Possible	RF interference
Sensor error	3	Minor	Possible	Noise
Visibility Loss	4	Significant	Possible	
Motor Overload	5	Significant	Unlikely	Excess acceleration
Power Failure	6	Severe	Unlikely	Battery undervoltage
Sensor Tolerances Too Large	7	Severe	Likely	Compounding error from sensors

Risk Matrix

Very Likely					
Likely					7
Possible		3		4	1,2
Unlikely				5	6
Very Unlikely					
	Negligible	Minor	Moderate	Significant	Severe

Mitigation Strategies



- Use XBee modules for reliable low-latency communication.
- Combine GPS + IMU + magnetometer using Kalman filtering.
- Ensure proper thermal venting and shade for ground components.
- Add protective logic for gimbal overload conditions.
- Perform magnetometer calibration away from ferrous material.
- Implement a lightweight computer vision model on the RPi or Jetson to minor corrections



Requirement Validation & Verification

This section will look back at the RFD and compare the specifications of the designed system to the requested system. Create a table which lists the given requirement, the relevant specification (if applicable), and whether the requirement was met/not met (verification). **This is a later section that you should reserve until most of the rest of your PDR is complete.**

- This is the place to describe how you will test your systems
- If you have a sensor, describe how you calibrate it. Look at resources we can find at tech such as the goodwin wind tunnel, etc.
- If you have a structural component, consider how you will test to ensure it won't break
- For example, the fuel tank should be tested by actually shaking it, putting it through a barrel roll, etc.
- Consider the final test of your entire system. Stress-test it, etc.

Consider including your own requirements to further define system success as well as requirements that may be given later such as bounding box, weight, and budget.

If the system is unable to meet a requirement, provide information on why – is it technically infeasible? Also consider whether the given requirements are reasonable in context of the entire mission. This is known as validation. If you feel a requirement is not valid. Provide information on why.

Requirement	Verification Method	Pass Criteria
180°/s azimuth	Bench rotation test	$\geq 180^\circ/\text{s}$
90°/s elevation	Bench rotation test	$\geq 90^\circ/\text{s}$
1° pointing accuracy	Laser alignment test	$\leq 1^\circ$ error
2 m position accuracy	Field GPS test	≤ 2 m
1.5-mile tracking	Outdoor range test	Maintains link
Orientation correction	Leveling calibration test	$\leq 1^\circ$ residual error
1 hour standby	Power drain test	≥ 60 min
20 min active	Continuous operation	≥ 20 min
Unwind routine <1 s	Spin test	<1 s recovery



Budget

System Cost

Item	Cost	Quantity
DJI Ronin-S	\$199	1
Digi XBee 3 PRO Zigbee 3.0	\$38.25	2
Triple-Axis Magnetometer LIS3MDL	\$9.95	2
ESP-WROOM-32 ESP32 ESP-32S Development Board	\$10.00	2
MAF94028: 4 Inch Omni Dipole Antenna	\$7.42	2
Adafruit Ultimate GPS Breakout	\$29.95	2
BMP390	\$10.95	1
Turnigy 2200mAh 3s 40C Lipo Pack	\$12	1
DFR1015 DC-DC Converter	\$5.30	1
Fujifilm X-T2	\$719.44	1
Adafruit LSM6DSO32 6=DoF Accelerometer and Gyroscope	\$12.50	1
LP803860 2000mAh 3.7V	\$12.50	2
Li-polymer Battery HAT, 5V Output, Quick Charge	\$18.99	1
Total Cost	Buffer 20%	
\$	1,194.32	955.456 ~ 1433.184
Actual Purchases		
DJI Ronin-S	\$199	1
Digi XBee 3 PRO Zigbee 3.0	\$38.25	2
Triple-Axis Magnetometer LIS3MDL	\$9.95	2
ESP-WROOM-32 ESP32 ESP-32S Development Board	\$10.00	2
MAF94028: 4 Inch Omni Dipole Antenna	\$7.42	2
LP803860 2000mAh 3.7V	\$12.50	2
Li-polymer Battery HAT, 5V Output, Quick Charge	\$18.99	1
Total Cost	Buffer 20%	
\$	374.23	299.384 ~ 449.076



Power Budget

External Module	Voltage	Amps (milli)	Watt Hours (milli)	Run Time	
Power Source	3.3	2200hr	7260	7.508532	Hr
EPS32	3.3	150	495		
LPS22DF	3.3	50	165		
GPS adafruit	3.3	53	174.9		
Digi Xbee	3.3	40	132		
Main System					
Power Source	3.7	3000	11100	0.827648	Hr
Digi Xbee	3.6	40	144		
RPI 4	5	2500	12500		
LIS3MDL	5	0.5	2.5		
LSM6DSOX	5	100	500		
GPS	5	53	265		
			13411.5		



Fall Schedule

Create a bullet point list of project milestones and their dates for the Fall semester. This will at least include when parts will be manufactured, testing deadlines, any internal reports, and final product delivery. There is great uncertainty in determining the project schedule thus changes are expected. This should be a reasonable schedule the project could continue to follow in the Fall to build and test your system. Your schedule should assume an initial Icarus flight on January 1st 2025 (for now).

Weeks 1–2: Design MVP and fill out POs

Weeks 3–4: Assemble external tracking module prototype

Weeks 5–6: Set up Gimbal with ROS2 and test commands

Weeks 7–8: Gimbal control algorithm implementation

Weeks 9–10: Full system integration and troubleshooting

Week 11: Fall PDR finalization and documentation submission

References

Use IEEE format for citations.